For a classical physicist this is very odd. No matter what you do, the energy always comes in unbreakable units.

You may already know that these units are called photons. In fact, *photon* is just another name for the quantized unit of energy in a quantum harmonic oscillator. But we can also describe the same facts another way. Being indivisible, photons can be thought of as elementary particles. A wave excited to its *n*th quantum state can be thought of as a collection of *n* photons.

What is the energy of a single photon? That's easy. It's just the energy that it takes to add one more unit, namely

$$E(\lambda) = \frac{2\pi\hbar c}{\lambda}.$$

Here, we can see something that has dominated physics for well over a century: the shorter the wavelength of a photon, the higher its energy. Why would a physicist be interested in making short-wavelength photons, given that they are costly in energy? The answer is to see more clearly. As discussed in Lecture 1, to resolve an object of a given size, you must use waves of that size or smaller. To see a human figure, a wavelength of a few inches is good enough. To see a tiny speck of dust, you may need visible light of a much smaller wavelength. To resolve the parts of a proton, the wavelength must be smaller than $10^{-15}$ meters, and the corresponding photons must be very energetic. In the end, it all goes back to the harmonic oscillator.

On that note, my friends, we conclude this volume of the Theoretical Minimum series. I look forward to seeing you in Special Relativity.